R. J. Tibshirani (2023)
Conformal Prediction (CMU Lecture Notes)

Paper Overview

1. Why this document exists

Ryan Tibshirani's 2023 CMU lecture notes Tibshirani (2023) are the statistician's introduction to conformal prediction (CP). Where Angelopoulos–Bates Angelopoulos and Bates (2022) optimise for ML-practitioner accessibility, Tibshirani teaches CP *as a statistician*, in the Vovk Vovk, Gammerman, and Shafer (2005) → Wasserman → Lei → Tibshirani CMU lineage.

Key Insight

Two flavours of intro: which to read. Angelopoulos–Bates is the ML-practitioner intro: it gives the recipe, the Python notebooks, the catalogue of scores you can swap in. Tibshirani 2023 is the statistician's intro: it derives the rank-based machinery from first principles, isolates the two "key ideas" that organise everything, and dedicates real space to the calibration-conditional Beta distribution and the Lei–Wasserman impossibility. Read both. The first builds confidence in deploying CP; the second builds confidence in extending it.

The notes are short (15 pages) and dense. They build the framework from first principles via two organising *key ideas*, derive split and full conformal as direct consequences, and then dedicate substantial space to the questions a statistician actually asks: how variable is the realised coverage on a single calibration set? What can we say about coverage conditional on the test input X? When can we trust the rank-based machinery and when does it fail?

Plain English

Why a statistician's view matters even if you're an engineer. The Angelopoulos–Bates tutorial gives you the recipe. The statistician's view tells you what *else* you should worry about: the variance of realised coverage on a single calibration set draw, the conditional-coverage impossibility, the distinction between symmetric and naive score construction. These are the things that bite in deployment but don't appear in the recipe.

This overview walks through the construction in the order Tibshirani presents it, makes the two key ideas explicit, and points at the connections to the rest of the wiki.

2. The two key ideas

The notes are organised around two ideas that, once isolated, make every subsequent construction in CP a corollary rather than a new invention.

Key idea 1: rank-based adjusted-level quantiles

Suppose we observe i.i.d. samples $Y_1, \dots, Y_n$ from some unknown distribution and want a prediction interval for a future observation Y_{n+1}. The standard textbook construction — empirical quantile of $Y_1, \dots, Y_n$ — under-covers at finite sample size because it treats Y_{n+1} as drawn from an estimated distribution rather than as a genuinely random new point. The fix is to take the quantile at level $\lceil (n+1)(1-\alpha) \rceil / n$ rather than $1-\alpha$:

$$\underbrace{\hat{q}}_{\substack{\text{adjusted-level}\\ \text{quantile}}} = \text{Quantile}\Bigg(\underbrace{\{Y_i\}_{i=1}^n}_{\text{calibration data}} ; \underbrace{\frac{\lceil (n+1)(1-\alpha) \rceil}{n}}_{\substack{\text{the +1 accounts for the}\\ \text{unseen test point}}} \Bigg).$$

Under exchangeability of $Y_1, \dots, Y_{n+1}$ — which i.i.d. already implies — the rank of Y_{n+1} among the $n+1$ observations is uniformly distributed on $\{1, \dots, n+1\}$. This single fact yields:

$$\underbrace{\mathbb{P}(Y_{n+1} \le \hat{q})}_{\substack{\text{probability test point}\\ \text{falls under cutoff}}} \ge \underbrace{1-\alpha}_{\text{nominal level}} .$$

The argument never uses any property of the distribution of Y — the guarantee is genuinely distribution-free at finite n. Auxiliary randomisation gives exact $1-\alpha$ coverage; without randomisation, the upper bound is $1-\alpha+1/(n+1)$ (the discreteness slack).

Plain English

The trick in one sentence. Use a quantile slightly above $1-\alpha$ to compensate for the fact that the test point is one of $n+1$ exchangeable observations, not the $(n+1)$-th draw from an estimated distribution. The whole CP framework is corollaries of this idea.

Key idea 2: symmetric score construction

Now suppose we have covariates X as well, and want a prediction set for Y_{n+1} given X_{n+1}. The naive approach is: fit $\hat{\mu}$ on training data, compute residuals $R_i = Y_i - \hat{\mu}(X_i)$ on the *same* training data, take the residual quantile, and form $[\hat{\mu}(X_{n+1}) \pm \hat{q}]$. This *undercovers* — often substantially — because the training-set residuals are smaller than they should be (the model has been fit to minimise them) while the test-set residual hasn't been.

Why This Matters

Why naive training-residual quantiles fail in practice. Take a flexible model (random forest, neural net) trained to small training error. Its training residuals are artificially small — the model has memorised. Quantiles of these residuals dramatically underestimate the test residuals' spread. A naive "± residual quantile" band built from training residuals can deliver 50% coverage when you asked for 90%. The fix is the symmetric-score discipline below.

The fix is to construct the score $s(X, Y)$ *symmetrically* with respect to the entire dataset including the test point. The standard implementation is the split-conformal trick: hold out a calibration set $\{(X_i, Y_i)\}_{i=1}^{n}$ from the training data so the residuals

$$\underbrace{R_i}_{\text{calibration residual}} = \underbrace{|Y_i - \hat{\mu}(X_i)|}_{\substack{\hat{\mu} \text{ never saw } (X_i, Y_i) \\ \text{during fitting}}}$$

are exchangeable with the (unobservable) test residual $R_{n+1} = |Y_{n+1} - \hat{\mu}(X_{n+1})|$. Now Key Idea 1 applies to $R_1, \ldots, R_n, R_{n+1}$, and the rank-based machinery delivers

$$\underbrace{\hat{C}(X_{n+1})}_{\text{prediction interval}} = \Big\{ y : \underbrace{|y - \hat{\mu}(X_{n+1})|}_{\text{score at candidate } y} \leq \underbrace{\hat{q}}_{\substack{\text{calibration} \\ \text{quantile}}} \Big\}.$$

Tibshirani's emphasis is that this is not a new construction — it is Key Idea 1 *applied to scores rather than raw observations.*

What This Definition Means

What "symmetric score" means. A score $s(\cdot,\cdot)$ is symmetric with respect to a dataset if its distribution doesn't change when you swap any data point with any other. Concretely: the score must treat the test point and any calibration point identically — neither can have been "used" more by the fitting procedure than the other. Split conformal achieves this by simply not letting $\hat{\mu}$ see calibration data.

3. From split to full to classification

The notes derive three families of conformal sets as variations on the same theme:

Split conformal. The construction above. One fit, fast, loses calibration data.

Full (transductive) conformal. For each candidate $y \in \mathbb{R}$, refit $\hat{\mu}$ on the augmented dataset $\{(X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, y)\}$, compute all $n+1$ residuals (including the augmented one), and include y in the set if its residual is among the smallest $\lceil (n+1)(1-\alpha) \rceil$. Costs $|\mathcal{Y}|$ refits per test point but loses no data. The notes give the elegant *p-value* interpretation: the conformal set is the set of y for which the augmented hypothesis "$Y_{n+1} = y$" would not be rejected at level α using a rank-based test.

How This Works

Full conformal as a hypothesis test. For each candidate y, ask: "if the true value were y, would the augmented dataset still look exchangeable?" The rank-based test answers this; the prediction set collects all the y values for which the answer is yes. This recasts CP as inverting a hypothesis test, in the classical Neyman–Pearson sense.

Classification. The construction extends to discrete $Y \in \{1, \ldots, K\}$ via either likelihood scores $s(x, y) = 1 - \hat{f}_y(x)$ (one minus the predicted softmax probability of the true class) or the cumulative-likelihood APS construction Sadinle, Lei, and Wasserman (2019). RAPS regularises APS to prevent pathological tail-class sets. Both inherit the rank-based coverage guarantee directly — the symmetric-score construction is what does the work, the underlying classifier is irrelevant.

4. Calibration-conditional coverage and the impossibility result

Tibshirani dedicates substantial space to two questions that go beyond the marginal-coverage guarantee.

Calibration-set-conditional coverage is Beta-distributed

The marginal coverage statement averages over both the test point *and* the calibration set draw. On a single fixed calibration set, the realised coverage C is itself a random variable, and Tibshirani derives its distribution explicitly:

$$\underbrace{C}_{\substack{\text{realised coverage on}\\ \text{this calibration set}}} \sim \mathrm{Beta}\Big(\underbrace{\lceil (n+1)(1-\alpha)\rceil}_{\text{shape 1}},\ \underbrace{n+1-\lceil (n+1)(1-\alpha)\rceil}_{\text{shape 2}}\Big).$$

This has practical consequences. The marginal coverage is $1-\alpha$ *averaged* over calibration draws, but the *variance* of C around that mean is non-negligible at small n. The rule of thumb that drops out: calibration set size $n \approx 1{,}000$ is required to keep realised coverage within ±1% of nominal with high probability. Smaller calibration sets give wider variability and — in the unlucky tail of the Beta — realised coverage substantially below nominal.

What This Result Says

What the Beta result really says for a deployed system. You typically have one calibration set in production, not many. The Beta distribution tells you exactly how much your realised coverage can drift away from nominal because of that single-set randomness. At $n = 100$, the standard deviation is $\sim 3\%$, so you could easily be running at 85% coverage when you think you're at 90%. At $n = 1000$, the standard deviation is $\sim 1\%$. Calibration-set size matters far more than most practitioners realise.

The X-conditional impossibility (Lei–Wasserman 2014)

The deeper question is whether the marginal guarantee can be strengthened to *conditional* coverage: $\mathbb{P}(Y \in \hat{C}(X) \mid X = x) \geq 1-\alpha$ for every x. Lei and Wasserman Lei and Wasserman (2014) proved this is impossible in the distribution-free setting at non-trivial level: any procedure that achieves conditional coverage uniformly over a non-atomic distribution must produce sets of infinite Lebesgue measure on a set of X of positive probability.

Why This Matters

Why this impossibility is the right negative result to internalise. If you want a per-individual coverage guarantee, you must give up either distribution-freeness (assume something about $P(Y|X)$) or non-triviality (allow infinite-measure sets, which means giving up). There is no middle ground in the worst case. Every "approximate conditional coverage" construction in CP (Mondrian, group-conditional, σ-scaled, CQR) is an honest engineering compromise around this hard fact, not a workaround.

Tibshirani's response is to reframe the practical goal as *local adaptivity*. The marginal-coverage guarantee says nothing about *where* the prediction set is wide or narrow; the locally-adaptive variants choose scores so that the set width tracks the conditional spread of $Y \mid X$. The studentised-residual variant (a special case of the σ-scaled construction of Lei et al. (2018)) and CQR Romano, Patterson, and Candès (2019) are the canonical regression solutions; APS/RAPS Sadinle, Lei, and Wasserman (2019) play the same role for classification by varying set size with input difficulty.

Key Insight

Local adaptivity as the practical substitute for conditional coverage. You can't get genuine per-x coverage distribution-free, but you can get *set widths that track the difficulty of each x*. A CQR interval at a hard x is wider than at an easy x; the marginal coverage is unchanged, but the per-x coverage is approximately equalised. "Local adaptivity" is the right thing to aim for in practice.

5. Limits, extensions, and connections

Tibshirani is careful to delineate what conformal prediction *does not* give you: any meaningful conditional guarantee (the impossibility theorem), validity under distribution shift (handled separately by weighted CP Tibshirani et al. (2019)), or validity under temporal dependence (handled by the non-exchangeable construction of Barber et al. (2023), the ACI updates discussed in Zaffran's work, or the EnbPI residual-refresh approach). Each of these limits is the entry point to a separate body of research, and the notes flag the literature as such rather than pretending the basic framework suffices.

Plain English

Where the simple framework stops working.

- Conditional coverage: ruled out by Lei–Wasserman in general; approximated by local adaptivity.
- Distribution shift: handled by weighted CP if shift is known, by ACI / NexCP / EnbPI otherwise.
- Time series: exchangeability fails; the four-family taxonomy of Stocker et al. organises the remedies.
- Missing values: CP-MDA (Zaffran et al.) restores mask-conditional validity.

Each item is a research literature in its own right.

The notes also clarify the relationship to other distribution-free prediction approaches: the Jackknife+ construction Barber et al. (2021) (distribution-free $1-2\alpha$ guarantee from leave-one-out residuals) is presented as the natural data-efficient alternative to split conformal, recovering most of the statistical efficiency of full CP at the cost of a single additional logical step (compare each LOO prediction to the test point's potential predictions). Non-exchangeable CP via fixed weights Barber et al. (2023) extends the same machinery to settings where the exchangeability of the underlying data fails outright.

For the wiki context: this paper is the statistician's-eye view of the same material covered in much greater applied detail by Angelopoulos and Bates. Read both. The statistician's emphasis on Beta-distributed calibration-conditional coverage, the symmetric-score discipline, and the X-conditional impossibility is missing from the practitioner tutorial and is exactly what a researcher building on CP needs to internalise.

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